VW2-DirectAct Push-T Continuous-Subgoal Falsification

Oracle-Fix Verification and Final Rerun

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Executive Summary. The rollout-side oracle-conditioning bug was fixed, regression tests were added, the TeacherOracle subgoal model was rerun, and the fixed evaluator was sanity-checked with the direct-act oracle action model. The final TeacherOracle result remains a decisive Gate A failure: **0.0% world success on execute-1 and 0.0% world success on execute-2**. The Push-T continuous-subgoal branch stops at Gate A.

Experiment ID. `pusht_subgoal_distill_round2_oraclefix`

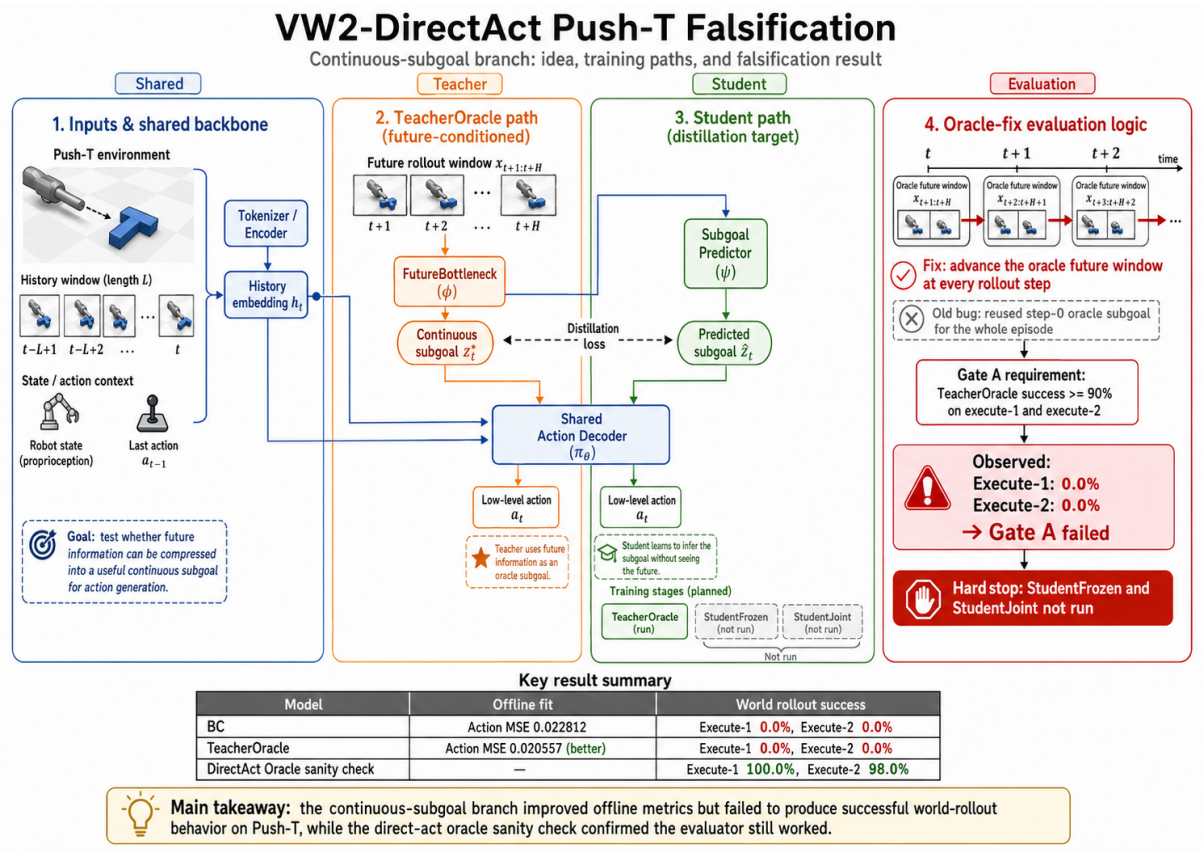

Figure 1: Overview illustration also used in the repository README. The committed source code, JSON/CSV artifacts, and this report remain the source of truth for metrics and conclusions.

1. Model Pipeline

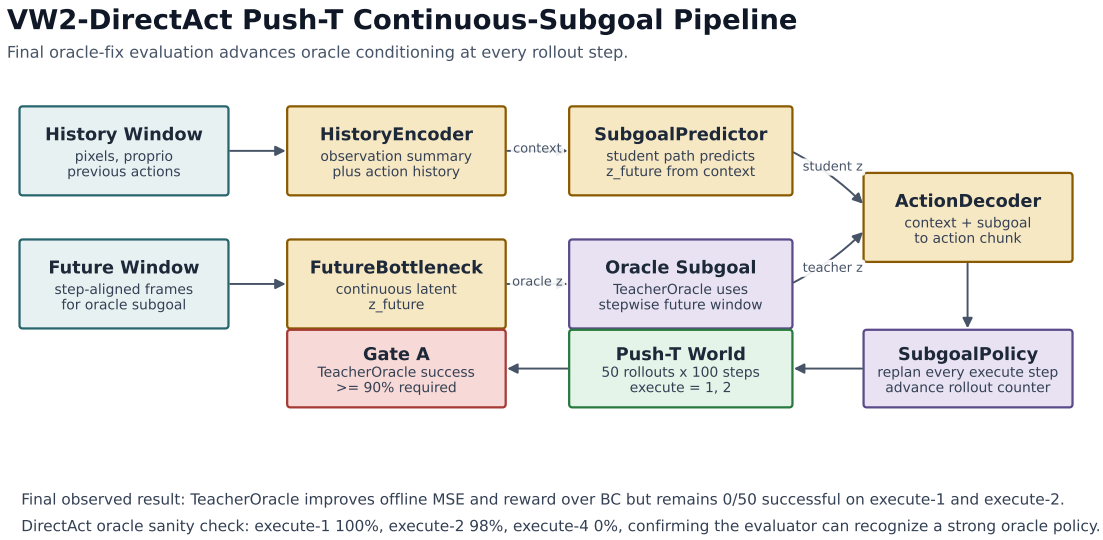

Figure 2: Continuous-subgoal pipeline used by the final Push-T oracle-fix rerun. The TeacherOracle path encodes a step-aligned future window at every rollout step.

The model uses a shared action decoder with two subgoal sources:

- **TeacherOracle path**: `FutureBottleneck` encodes a future rollout window into a continuous subgoal.
- **Student path**: `SubgoalPredictor` predicts the continuous subgoal from encoded history.

The final rerun evaluates the TeacherOracle path. StudentFrozen and StudentJoint are intentionally skipped after Gate A fails.

2. Experiment Flow

Gate A requires TeacherOracle success of at least 90% on execute-1 and execute-2 over 50 rollouts with 100 world steps per rollout. The final rerun uses `eval.rollout_batch_size=10` for stable local RTX 4070 8 GB evaluation. This changes batching only, not evaluation semantics.

Evaluation scope. The world rollouts are deterministic starts from `pusht_expert_train.h5`, selected by the evaluator from valid expert episodes. They are not an episode-held-out test split. BC and the direct-act oracle sanity check use direct-act evaluator starts: episodes 0–49 at `start_step=0`. TeacherOracle uses subgoal evaluator starts: episodes 101–150 at `start_step=3`. Offline train/validation samples are window-level splits, so offline validation metrics should not be interpreted as episode-level generalization evidence.

3. Implementation Fixes

- **Stepwise oracle conditioning.** `DirectActPolicy` and `SubgoalPolicy` consume per-step oracle plan/subgoal tensors and advance a rollout-step counter during world evaluation.

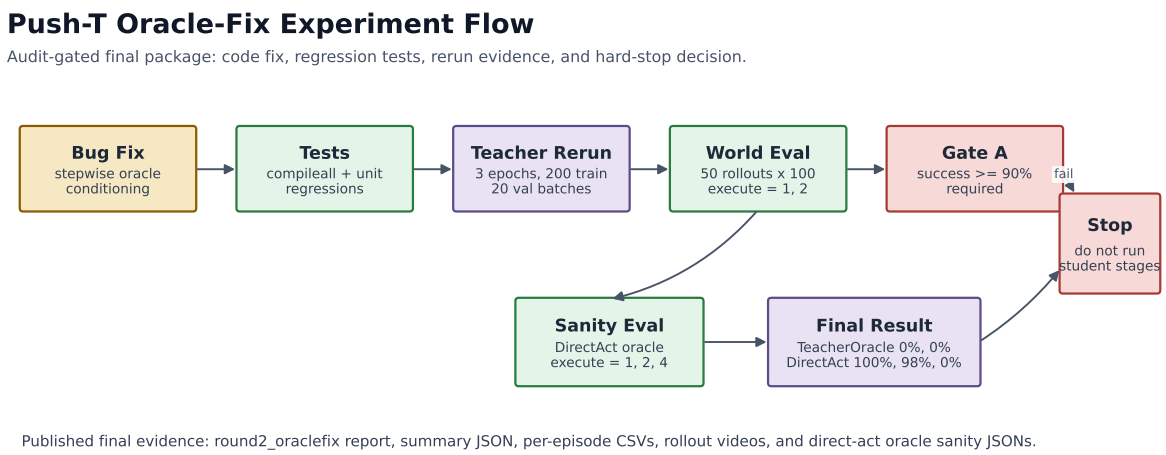

Figure 3: Final experiment flow from oracle-conditioning fix to Gate A hard stop.

- **TeacherOracle world evaluation.** `eval_subgoal_policy.py` generates oracle subgoals from rollout windows aligned to the current rollout step.
- **Direct-act oracle evaluation.** `eval_policy.py` advances oracle plan conditioning by rollout step and uses the configured maximum evaluation length.
- **Rollout count guard.** World evaluation now raises an error if fewer valid starts are found than `eval.num_rollouts` requests.
- **Regression coverage.** Unit tests verify stepwise oracle plan selection and stepwise oracle subgoal selection.

4. Training Setup

Item	Value
Teacher stage	`train_teacher_oracle.py`
Epochs	3
Train batches per epoch	200
Validation batches per epoch	20
Train samples	12000
Validation samples	1024
Future horizon	8
History steps	4
Action chunk	4
Eval rollouts	50
Eval steps	100
Execute sweep	$\{1, 2\}$

Final TeacherOracle training metrics from the published CSV log:

Metric	Value
`val/loss`	0.0108936075
`val/teacher_action_mse`	0.0217941180
`val/subgoal_batch_variance`	0.0000237405
`val/covariance_offdiag_mean`	0.0000103518

The offline action fit is numerically strong, while the subgoal representation has very low batch variance. This is consistent with a bottleneck that does not encode enough closed-loop control information for Push-T.

5. World Results

Mode	MSE	Exec-1	Reward-1	Exec-2	Reward-2
BC	0.02281195	0.0%	-23644.98	0.0%	-24199.06
TeacherOracle	0.02055714	0.0%	-21435.20	0.0%	-20406.16

TeacherOracle improves offline action MSE and mean reward over BC, yet neither policy achieves a single success in 50 rollouts on execute-1 or execute-2.

Evaluator sanity check. The same fixed evaluator was rerun on the direct-act oracle action model:

Mode	Exec-1 Success	Exec-2 Success	Exec-4 Success
DirectAct Oracle	100.0%	98.0%	0.0%

This confirms that the evaluator can still recognize a strong oracle policy after the fix. The continuous-subgoal result is a branch failure, not a confirmed evaluator failure.

6. Gate Decision

Gate	Rule	Result
Gate A	TeacherOracle success $\geq 90\%$ on execute-1 and execute-2	**FAIL**
Hard stop	If Gate A fails, stop immediately and skip StudentFrozen and StudentJoint	**TRIGGERED**

7. Public Artifacts

- Final artifact root: `artifacts/pusht_subgoal_distill_round2_oraclefix/`
- Main summary: `eval_subgoal_50rollouts_100steps/summary.json`
- Per-episode CSVs: BC and TeacherOracle folders in the subgoal evaluation directory
- Direct-act oracle sanity videos: `eval_50rollouts_100steps/videos_execute_1/`
- TeacherOracle videos: `videos_execute_1/` under the TeacherOracle subgoal folder
- Direct-act oracle sanity JSONs: `directact_oracle_eval_50rollouts_100steps/`

- Overview figure shared with README: `pusht_falsification_overview.png`
- Architecture diagrams: `model_pipeline.svg` and `experiment_flow.svg`

Checkpoints and raw datasets are intentionally excluded from version control.